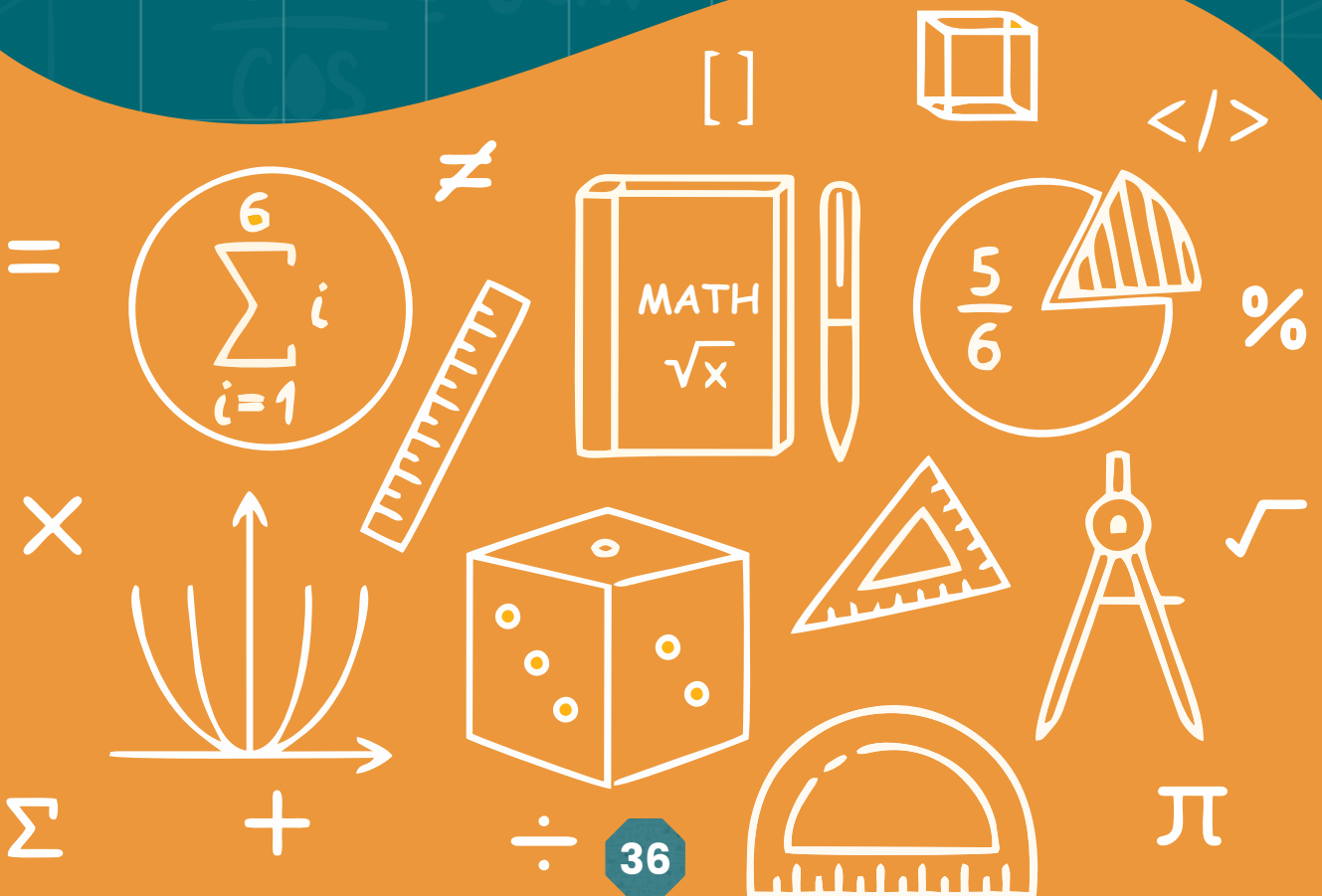


UNIT 4

EXPONENTS



EXPONENTS/ INDICES

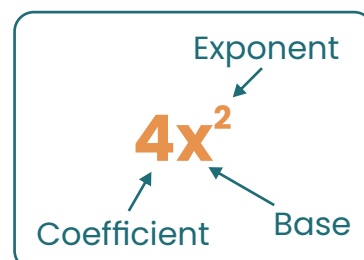
An exponent is a small number written above and to the right of a base number. It tells you how many times to multiply the base number by itself

Base:

Base is a number which is to be multiplied by itself according to exponent.

Exponent or Index:

Exponent is a number which represents the number of times, a base is to be multiplied by itself.



How to read Exponents:

2^3 is read as "2 to the power of 3"

5^2 is read as "5 to the power of 2"

To calculate an exponent, you multiply the base number by itself as many times as the exponent says.

For example:

$$2^3 = 2 \times 2 \times 2 = 8$$

$$5^2 = 5 \times 5 = 25$$

Rules of Exponents

- Any number raised to the power of 0 is equal to 1
(e.g., $2^0 = 1$)
- Any number raised to the power of 1 is equal to itself
(e.g., $2^1 = 2$)

$$a^0 = 1$$

Example 1: Identify Base and Exponent

Identify base and exponent in the following. Also, find the value.

i. 2^4 ii. 5^2 iii. $\left(\frac{4}{7}\right)^3$ iv. $\left(\frac{2}{3}\right)^4$

S. No.	Expression	Base	Exponent	Value
1	2^4	2	4	16 because $2 \times 2 \times 2 \times 2 = 16$
2	5^2	5	2	25 because $5 \times 5 = 25$

3	$\left(\frac{4}{7}\right)^3$	$\frac{4}{7}$	3	$\frac{64}{343}$ because $\frac{4}{7} \times \frac{4}{7} \times \frac{4}{7} = \frac{64}{343}$
4	$\left(\frac{2}{3}\right)^4$	$\frac{2}{3}$	4	$\frac{16}{81}$ because $\frac{2}{3} \times \frac{2}{3} \times \frac{2}{3} \times \frac{2}{3} = \frac{16}{81}$

LAWS OF EXPONENTS

Product Rule:	$a^m \times a^n = a^{(m+n)}$
Quotient Rule:	$a^m / a^n = a^{(m-n)}$
Power of a Power Rule:	$(a^m)^n = a^{(m \times n)}$
Power of a Product Rule:	$(ab)^m = a^m \times b^m$
Power of a Quotient Rule:	$(a/b)^m = a^m / b^m$
Zero Exponent Rule:	$a^0 = 1$
Negative Exponent Rule:	$a^{(-m)} = 1/a^m$
Fractional Exponent Rule:	$a^{(m/n)} = \sqrt[n]{a^m}$

Law	Example
$a^m \times a^n = a^{(m+n)}$	$2^3 \times 2^4 = 2^{(3+4)} = 2^7 = 128$
$(a^m)^n = a^{(m \times n)}$	$(2^3)^4 = 2^{(3 \times 4)} = 2^{12} = 4096$
$(ab)^n = a^n \times b^n$	$(20)^3 = (2 \times 10)^3 = 2^3 \times 10^3 = 8000$
$(a/b)^m = a^m / b^m$	$(2^3 / 5^3) = 8 / 125$
$a^m / a^n = a^{(m-n)}$	$2^5 / 2^3 = 2^{(5-3)} = 2^2 = 4$
$1/a^n = a^{(-n)}$	$1/2^3 = 2^{(-3)} = 1/8$

Example:

$$2^3 \times 2^2 = (2 \times 2 \times 2) \times (2 \times 2) \\ = 2 \times 2 \times 2 \times 2 \times 2 = 2^5$$

i.e., $2^3 \times 2^2 = 2^{(3+2)} = 2^5$

Example: Simplify using Product Law

i. $\left(\frac{2}{7}\right)^4 \times \left(\frac{2}{7}\right)^5$ ii. $\left(\frac{3}{4}\right)^5 \times \left(\frac{3}{4}\right)^6 \times \left(\frac{2}{5}\right)^4 \times \left(\frac{2}{5}\right)^5$

Solution:

i. $\left(\frac{2}{7}\right)^4 \times \left(\frac{2}{7}\right)^5$

$$= \left(\frac{2}{7}\right)^{4+5} \text{ (using product law)}$$

$$= \left(\frac{2}{7}\right)^9$$

ii. $\left(\frac{3}{4}\right)^5 \times \left(\frac{3}{4}\right)^6 \times \left(\frac{2}{5}\right)^4 \times \left(\frac{2}{5}\right)^5$

$$= \left(\frac{3}{4}\right)^{5+6} \times \left(\frac{2}{5}\right)^{4+5} \text{ (using product law)}$$

$$= \left(\frac{3}{4}\right)^{11} \times \left(\frac{2}{5}\right)^9$$

Example:

In order to deduce this law, let us consider the following examples.

$$(7^3)^2 = (7^3) \times (7^3) \\ = (7 \times 7 \times 7) \times (7 \times 7 \times 7) = 7^6$$

i.e., $(7^3)^2 = 7^{(3 \times 2)}$

Similarly,

$$\left\{\left(\frac{11}{12}\right)^2\right\}^4 = \left(\frac{11}{2}\right)^2 \times \left(\frac{11}{2}\right)^2 \times \left(\frac{11}{2}\right)^2 \times \left(\frac{11}{2}\right)^2$$

$$= \left(\frac{11}{2} \times \frac{11}{2}\right) \times \left(\frac{11}{2} \times \frac{11}{2}\right) \times \left(\frac{11}{2} \times \frac{11}{2}\right) \times \left(\frac{11}{2} \times \frac{11}{2}\right) \times \left(\frac{11}{2}\right)^8$$

i.e., $\left\{\left(\frac{11}{12}\right)^2\right\}^4 = \left(\frac{11}{2}\right)^8 = \left(\frac{11}{2}\right)^{2 \times 4}$

Example: Which of the following are positive or negative?

- (i) $(-5)^{11}$ (ii) $(-3)^{20}$ (iii) $(-8)^{113}$ (iv) $(-5)^{40}$
 (v) $(-8)^0$ (vi) $(-7)^{-5}$ (vii) $(-110)^{-40}$

Solution:

S. No.	Expression	Positive or Negative	Expression
1	$(-5)^{11}$	Negative	The exponent is odd
2	$(-3)^{20}$	Positive	The exponent is even
3	$(-8)^{113}$	Negative	The exponent is odd
4	$(-5)^{40}$	Positive	The exponent is even
5	$(-8)^0$	Positive	The exponent is even
6	$(-7)^{-5}$	Negative	The exponent is odd
7	$(-110)^{-40}$	Positive	The exponent is even

EXERCISE 4.1

1. Expand and find the value of each exponential form.

(i) 3^2

(ii) 2^7

(iii) $2^3 \times 3^4$

(iv) $5^2 \times 7^2$

(v) $2^3 \times 3^2 \times 5^2$

(vi) $2^4 \times 3^3 \times 7^2$

2. Identify the base and exponent. Also find the value of each of the following expressions.

i. 2^4

ii. 5^4

iii. 6^3

iv. $\left(\frac{2}{3}\right)^6$

v. $\left(\frac{-1}{5}\right)^5$

3. Simplify using product law.

i. $\left(\frac{5}{7}\right)^2 \times \left(\frac{5}{7}\right)^4$

ii. $\left(\frac{3}{5}\right)^6 \times \left(\frac{1}{5}\right)^6$

iii. $\left(\frac{1}{7}\right)^5 \times \left(\frac{1}{7}\right)^5 \times \left(\frac{-5}{7}\right)^8 \times \left(\frac{-5}{7}\right)^8$

iv. $p^6 \times p^7$

v. $y^5 \times x^{11} \times y^6$

vi. $\left(\frac{4}{3}\right)^5 \times \left(\frac{4}{3}\right)^6 \times \left(\frac{4}{3}\right)^7$

4. Verify that:

(i) $2^5 \times 3^5 = 6^5$

(ii) $p^7 \times q^7 = (pq)^7$

(iii) $\left(\frac{3}{4}\right)^4 \times \left(\frac{3}{4}\right)^8 = \left(\frac{3}{4}\right)^{12}$

5. Simplify:

(i) $5^7 \div 5^4$

(ii) $\left(\frac{1}{2}\right)^6 \div \left(\frac{1}{2}\right)^2$

(iii) $x^{10} \div x^4$

(iv) $(5)^6 \div (3)^6$

(v) $(10)^6 \div (2)^5$

6. Verify:

(i) $\left(\frac{2}{9}\right)^5 \div \left(\frac{2}{9}\right)^2 = 2^3 \div 9^3$

(ii) $(-3)^8 \div (-5)^8 = \left(\frac{3}{5}\right)^{10} \div \left(\frac{3}{5}\right)^2$

(iii) $\left(\frac{x}{y}\right)^7 \div \left(\frac{x}{y}\right)^5 = x^3 \div y^3$

(iv) $5^{11} \div 5^8 = (20)^3 \div (4)^3$

7. Simplify using power law.

(i) $(52)^7$

(ii) $\left\{\left(\frac{1}{3}\right)^6\right\}^8$

(iii) $((-3)^2)^{10}$

(iv) $(x^5)^{20}$

(v) $\left\{\left(\frac{4}{7}\right)^9\right\}^2$

(vi) $\left\{\left(\frac{-3}{4}\right)^{20}\right\}^2$

8. Convert into negative exponent.

(i) $\frac{1}{2^5}$

(ii) $\frac{1}{3^7}$

(iii) $\frac{1}{x^6}$

(iv) $\frac{1}{\left(\frac{4}{5}\right)^6}$

9. Convert into positive exponent.

(i) 5^{-8}

(ii) $\left(\frac{1}{3}\right)^{-7}$

(iii) x^{-5}

(iv) y^{-7}

10. Simplify:

(i) $\left(\frac{5}{7}\right)^6 \times \left(\frac{5}{7}\right)^4 \times \left(\frac{5}{7}\right)^{10}$

(ii) $\left\{\left(\frac{2}{3}\right)^2\right\}^3 \div \left(\frac{2}{3}\right)^4$

(iii) $\left(\frac{7}{9}\right)^3 \times \left(\frac{9}{7}\right)^3 + \left(\frac{11}{12}\right)^5 \times \left(\frac{12}{11}\right)^5$

(iv) $x^0 + x^5 \times x^{-5}$